

Section : Chemistry

Q.1 The factor(s) governing the flow of electricity through an electrolytic solution is/are

- (A) The nature of inter-ionic attraction.
- (B) Temperature
- (C) Solvation of ions.
- (D) The nature of solvent and its viscosity.

Choose the correct answer from the options given below:

- 1. (D) only
- 2. (A) and (C) only
- 3. (A), (B), (C) and (D)
- 4. (B), (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.2 What is the symbol of technetium element ?

- 1. Ta
- 2. Tc
- 3. Ti
- 4. Th

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.3 Kohlrausch's law is used for the determination of

- (A). Molar conductivity of a weak electrolyte.
- (B). Degree of dissociation of a weak electrolyte.
- (C). Dissociation constant.
- (D). Solubility of sparingly soluble salts.

Choose the correct answer from the options given below:

- 1. (A) and (B) only
- 2. (A) and (C) only
- 3. (A), (B), (C) and (D)
- 4. (B), (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.4 What is the coordination number of Ni in $[\text{Ni}(\text{NH}_3)_4]$ complex?

- 1. 7
- 2. 3
- 3. 12
- 4. 4

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.5 On doubling the initial concentration, the half life of a reaction becomes half. The order of the reaction is

- 1. 0.5
- 2. 0
- 3. 2
- 4. 1

Options 1. 1

- 2. 2
- 3. 3

Q.6 What product(s) is (are) formed when an alkyl halide undergoes ammonolysis ?

- (A) Primary amine
- (B) Secondary amine
- (C) Quaternary ammonium salt
- (D) Tertiary amine

Choose the correct answer from the options given below:

- 1. (A), (B) and (C) only
- 2. (A), (B), (C) and (D)
- 3. (B), (C) and (D) only
- 4. (A), (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.7 The zwitter ion formed in an aqueous solution of glycine is

- 1. Neutral
- 2. Acidic
- 3. Basic
- 4. Asymmetric

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.8 What is the product formed when benzyl ethyl ether reacts with HI?

- 1. Benzyl iodide and methyl alcohol
- 2. Benzyl iodide and propyl alcohol
- 3. Benzyl iodide and butyl alcohol
- 4. Benzyl iodide and ethyl alcohol

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.9 Name the reaction by which Freons 12 are manufactured

- 1. Wurtz-Fittig reaction
- 2. Swarts reaction
- 3. Electrophilic substitution reaction
- 4. Free radical halogenation

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.10 Which of the following vitamins is soluble in water?

- 1. Vitamin E
- 2. Vitamin D
- 3. Vitamin B
- 4. Vitamin K

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.11 What are the two monosaccharides which are obtained by hydrolysis of lactose sugar?

- 1. D-(+)-Fructose and D-(+)-Galactose
- 2. D-(+)-Maltose and D-(+)-Galactose
- 3. D-(+)-Glucose and D-(+)-Galactose
- 4. D-(+)-Maltose and D-(+)-Fructose

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.12 Match List-I with List-II

List-I (Compound)

(A) Diazonium salt

(B) Novocain

(C) Benadryl

(D) Quaternary ammonium salts

List-II (Uses)

(I) Surfactants

(II) Antihistaminic drug

(III) Dentistry

(IV) Dyes

Choose the correct answer from the options given below:

- 1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- 2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
- 3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- 4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.13 Match List-I with List-II

List-I (Name of the Reaction)

(A) Kolbe's Reaction

(B) Reimer-Tiemann Reaction

(C) Etard Reaction

(D) Sandmeyer Reaction

List-II (Compound synthesized)

(I) Salicylaldehyde

(II) Salicylic acid

(III) Chlorobenzene

(IV) Benzaldehyde

Choose the correct answer from the options given below:

- 1. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- 2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- 3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
- 4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.14 What is the increasing order of reactivity of the following compounds towards S_N1 reactions?

(A). $C_6H_5CH_2Br$

(B). $C_6H_5CH(C_6H_5)Br$

(C). $C_6H_5C(CH_3)(C_6H_5)Br$

(D). $C_6H_5CH(CH_3)Br$

Choose the correct answer from the options given below:

- 1. (A), (B), (C), (D)
- 2. (A), (C), (B), (D)
- 3. (B), (A), (D), (C)
- 4. (A), (D), (B), (C)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.15 What is the IUPAC name of allylamine?

- 1. Prop-2-en-1-amine
- 2. Aniline
- 3. 2-methylaniline
- 4. Propan-2-amine

Options 1. 1

2. 2
3. 3
4. 4

Q.16 What is the correct IUPAC name of $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

1. diamminedichloridoplatinum (II)
2. diamminedichloridoplatinum (IV)
3. diamminedichloridoplatinum (0)
4. dichloridodiammineplatinum (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.17 What is the IUPAC name of the product of hydration of propene in the presence of dilute sulphuric acid?

1. 2-Methylpropan-2-ol
2. Propan-1-ol
3. Propan-2-ol
4. 2-Methylbutan-2-ol

Options 1. 1

2. 2
3. 3
4. 4

Q.18 Arrange the following complexes in decreasing order of number of unpaired electrons on the metal ion

- (A) $[\text{Co}(\text{NH}_3)_6]^{3+}$
(B) $[\text{CoF}_6]^{3-}$
(C) $[\text{FeF}_6]^{3-}$
(D) $[\text{Fe}(\text{CN})_6]^{3-}$

Choose the correct answer from the options given below:

1. (A), (D), (B), (C)
2. (A), (D), (C), (B)
3. (C), (B), (D), (A)
4. (C), (D), (A), (B)

Options 1. 1

2. 2
3. 3
4. 4

Q.19 Arrange the following alkyl halides in increasing order of boiling points

- (A) 1-Iodobutane
(B) 1-Bromobutane
(C) 1-Chlorobutane
(D) Butane

Choose the correct answer from the options given below:

1. (D), (C), (B), (A)
2. (A), (B), (C), (D)
3. (B), (C), (A), (D)
4. (A), (C), (B), (D)

Options 1. 1

2. 2
3. 3
4. 4

Q.20 The Galvanic cells will behave like an electrolytic cell when

1. $E_{\text{cell}} = 0$
2. $E_{\text{cell}} > E_{\text{ext}}$

3. $E_{\text{cell}} < E_{\text{ext}}$

4. $E_{\text{cell}} = E_{\text{ext}}$

Options 1. 1

2. 2

3. 3

4. 4

Q.21 What will happen when the temperature of a chemical reaction rises by 10°C

1. rate constant remains same.
2. rate constant is nearly doubled.
3. rate constant decreases.
4. rate constant increases by 10 times.

Options 1. 1

2. 2

3. 3

4. 4

Q.22 Coupling of benzene diazonium chloride with phenol yields colored compounds called azo dyes. The reaction requires

1. acidic medium
2. alkaline medium
3. neutral medium
4. catalyst

Options 1. 1

2. 2

3. 3

4. 4

Q.23 Among the following, which is a non-essential amino acid?

1. Arginine
2. Lysine
3. Tyrosine
4. Histidine

Options 1. 1

2. 2

3. 3

4. 4

Q.24 Match the compounds given in List-I with their application given in List-II

List-I (compound) List-II (Applications)

- | | |
|----------------------|-----------------------------|
| (A). TiO_2 | (I). dry battery cells |
| (B). AgBr | (II). Wacker process |
| (C). MnO_2 | (III). pigment industry |
| (D). PdCl_2 | (IV). photographic industry |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (I), (B) - (IV), (C) - (III), (D) - (II)
3. (A) - (III), (B) - (I), (C) - (II), (D) - (IV)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.25 Arrange the following in increasing order of the mass of the metal deposited when 1 amp of current is passed through their electrolytic solutions for 1 minute

- (A). Al (atomic mass = 27 g/mol)
- (B). Mg (atomic mass = 24 g/mol)
- (C). Ag (atomic mass = 108 g/mol)
- (D). Cu (atomic mass = 63.5 g/mol)

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (B), (D), (C)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.26 Select the correct statements about lanthanoids:

- (A). Lanthanoids are good conductors of heat and electricity
- (B). Ln oxides are used as phosphors in TV screens
- (C). Their Melting points range between 2100 to 2300 K
- (D). It is used for the production of alloy steels

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (C) and (D) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.27 How many moles of dichromate ion will be required to oxidize one mole of Fe^{2+} to Fe^{3+} ions in acidic solution?

1. 3
2. 6
3. $1/3$
4. $1/6$

Options 1. 1

2. 2
3. 3
4. 4

Q.28 Which of the following has the highest molar conductivity?

1. potassium hexacyanoferrate(II)
2. hexaaquachromium(II)chloride
3. tetraamminedichloridocobalt(III)chloride
4. potassium tetracyanonickelate(II)

Options 1. 1

2. 2
3. 3
4. 4

Q.29 What happens when a tertiary alcohol is heated at temperature of 443 K in presence of phosphoric acid?

1. Ether is formed
2. Alkene is formed
3. Mixture of alkene and alcohol is formed
4. No reaction takes place

Options 1. 1

2. 2
3. 3
4. 4

Q.30 Which species is involved in chlorination of benzene using Cl_2 in presence of AlCl_3 ?

1. Cl^+
2. $\text{Cl}^\cdot$
3. AlCl_4^+

4. AlCl_4^-

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.31 Nitriles on reduction with lithium aluminum hydride produce

- 1. Amides
- 2. Primary amine
- 3. Ammonium salt
- 4. Secondary amine

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.32 Match the type of ligands given in List-I with appropriate examples given in List-II

List-I (Ligands) List-II (Examples)

- (A). Ambidentate (I). thiocyanato
- (B). Unidentate (II). water
- (C). Didentate (III). oxalate
- (D). Polydentate (IV). ethylenediaminetetraacetate

Choose the correct answer from the options given below:

- 1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- 2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
- 3. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)
- 4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.33 Rusting of Iron is catalyzed by:

- 1. H^+
- 2. O_2
- 3. Fe
- 4. CO_3^{2-}

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.34 An oxidation of phenol with chromic acid produces

- 1. Benzoic acid
- 2. Salicylic acid
- 3. 2,4,6-Tribromophenol
- 4. Benzoquinone

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.35 In the following examples, select the primary alkyl halide

- (A). 1-Bromo-3-methylbutane
- (B). 2-Bromo-2-methylbutane
- (C). 3-Bromo propene
- (D). Chlorophenyl methane

Choose the correct answer from the options given below:

- 1. (A), (C) and (D) only
- 2. (A), (B) and (D) only

3. (A), (B) and (C) only

4. (A), (B), (C) and (D)

Options 1. 1

2. 2

3. 3

4. 4

Q.36 What is a one letter code of Tryptophan?

1. T

2. Y

3. I

4. W

Options 1. 1

2. 2

3. 3

4. 4

Q.37 From the statements given, choose the correct option based on the Arrhenius equation

$$k = Ae^{-E_a/RT}$$

1. rate constant increases exponentially with increasing activation energy and decreasing temperature.

2. rate constant decreases exponentially with increasing activation energy and decreasing temperature.

3. rate constant increases exponentially with decreasing activation energy and decreasing temperature.

4. rate constant increases exponentially with decreasing activation energy and increasing temperature.

Options 1. 1

2. 2

3. 3

4. 4

Q.38 The correct mathematical representation of the rate of a reaction is

1. Arrhenius equation

2. Collision frequency

3. Molecularity of a reaction

4. Rate law equation

Options 1. 1

2. 2

3. 3

4. 4

Q.39 What is the increasing order of the oxidation state of the metal ion in following compounds?

(A).ScBr₃

(B).V₂O₅

(C).CrF₆

(D).Tc₂O₇

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)

2. (A), (C), (B), (D)

3. (B), (A), (D), (C)

4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.40 Match the items given in List-I with items given in List-II

List-I

(A) Mathematical expression for the rate of reaction is

(B) The rate of reaction for zero order reaction is equal to

(C) The unit of rate constant for zero order reaction is the same as that of (III) rate constant

(D) The rate of a complex reaction is determined by

List-II

(I) rate of slowest step

(II) rate law

(IV) rate of a reaction

Choose the correct answer from the options given below:

- (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
- (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
- (A)-(III), (B)-(II), (C)-(IV), (D)-(I)
- (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

Options 1. 1

2. 2

3. 3

4. 4

Q.41 Read the passage carefully and answer the questions

The osmotic pressure is one of the four colligative properties which depends on the number of particles of solute in the solution, irrespective of their nature. The osmotic pressure is equal to the product of concentration, gas constant and temperature. It is the most commonly used property for the determination of the molar mass of biomolecules. In many situations, the molecules get dissociated or associated when they are dissolved in a solvent, thereby changing the number of molecules in the solution. As a result, an abnormal molecular mass is obtained. The extent of dissociation or association is given by van't Hoff factor (i), which is the ratio of normal molar mass to abnormal molar mass. It is related with the degree of association or ionization. The introduction of van't Hoff factor in the osmotic pressure equation modifies the equation.

$$\pi = i \cdot C \cdot R \cdot T$$

where π is the osmotic pressure, C is the concentration, R is the gas constant, T is the temperature, and i is the van't Hoff factor.

1. Number of solute molecules.
2. Temperature.
3. Concentration of solution.
4. Nature of solute particles.

Options 1. 1

2. 2

3. 3

4. 4

Q.42 Read the passage carefully and answer the questions

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The osmotic pressure of a 0.1 M aqueous solution of K_2SO_4 is 4.92 atm at 300 K. The percentage ionization of the salt is: Given $R = 0.082 \text{ atm L K}^{-1} \text{ mol}^{-1}$

1. 50
2. 39
3. 80
4. 10

Options 1. 1

2. 2

3. 3

4. 4

Q.43 Read the passage carefully and answer the questions

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associated when they are dissolved in a solvent, thereby changing the number of molecules in the solution. As a result, an abnormal molecular mass is obtained. The extent of dissociation or association is given by van't Hoff factor (i), which is the ratio of normal molar mass to abnormal molar mass. It is related with the degree of association or ionization. The introduction of van't Hoff factor in the osmotic pressure equation modifies the equation.

How is the osmotic pressure of a concentrated solution related to the osmotic pressure of a dilute solution of the same substance at a given temperature?

1. Lower than that of its dilute solution.
2. Same as its dilute solution.
3. Higher than that of its dilute solution.
4. Cannot compare on concentration basis until we know the nature of that solute.

Options 1. 1

2. 2
3. 3
4. 4

Q.44 Read the passage carefully and answer the questions

The osmotic pressure is one of the four colligative properties which depends on the number of particles of solute in the solution, irrespective of their nature. The osmotic pressure is equal to the product of concentration, gas constant and temperature. It is the most commonly used property for the determination of the molar mass of biomolecules. In many situations, the molecules get dissociated or associated when they are dissolved in a solvent, thereby changing the number of molecules in the solution. As a result, an abnormal molecular mass is obtained. The extent of dissociation or association is given by van't Hoff factor (i), which is the ratio of normal molar mass to abnormal molar mass. It is related with the degree of association or ionization. The introduction of van't Hoff factor in the osmotic pressure equation modifies the equation.

What is the value of the van't Hoff factor for NaCl , K_2SO_4 and KCl , respectively?

1. 1, 3 and 1
2. 1, 2 and 1
3. 1, 2 and 3
4. 2, 3 and 2

Options 1. 1

2. 2
3. 3
4. 4

Q.45 Read the passage carefully and answer the questions

The osmotic pressure is one of the four colligative properties which depends on the number of particles of solute in the solution, irrespective of their nature. The osmotic pressure is equal to the product of concentration, gas constant and temperature. It is the most commonly used property for the determination of the molar mass of biomolecules. In many situations, the molecules get dissociated or associated when they are dissolved in a solvent, thereby changing the number of molecules in the solution. As a result, an abnormal molecular mass is obtained. The extent of dissociation or association is given by van't Hoff factor (i), which is the ratio of normal molar mass to abnormal molar mass. It is related with the degree of association or ionization. The introduction of van't Hoff factor in the osmotic pressure equation modifies the equation.

Which one of the following has minimum osmotic pressure at 298 K?

1. 3 M KCl solution
2. 1 M glucose solution.
3. 2 M urea solution.
4. 2 M K_2SO_4 solution

Options 1. 1

2. 2
3. 3
4. 4

Q.46 Read the passage carefully and answer the Questions

Carboxylic acids are the earliest organic compounds isolated from nature. These are also known as organic acids but are weaker acids as compared to mineral acids. However, they are more acidic than alcohols and phenols because of greater resonance stabilization of the parent acid and its conjugate base. The acidic strength is determined by the K_a , which is called the acidity constant.

The greater the value of K_a , the greater the tendency of the acid to ionize and hence the stronger the acid. The acidic strength of the saturated aliphatic carboxylic acid depends upon the inductive effect and its position with respect to $-\text{COOH}$ groups. Electron -donating substituents tend to decrease whereas electron -withdrawing substituents tend to increase the acidic strength. The acidic strength of the aromatic carboxylic acid, on the other hand depends upon both the inductive effect and the resonance effect of the substituents.

Arrange the following acids in decreasing order of acidity

- (A) $\text{CH}_3\text{CH}_2\text{OH}$
- (B) CH_3COOH
- (C) BrCH_2COOH
- (D) $\text{C}_6\text{H}_5\text{CH}_2\text{COOH}$

Choose the correct answer from the options given below:

- 1. (A), (B), (C), (D)
- 2. (C), (D), (B), (A)
- 3. (B), (A), (D), (C)
- 4. (C), (B), (D), (A)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.47 Read the passage carefully and answer the Questions

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Which of the following groups when attached directly does not increase the acidity of the carboxylic acid ?

- 1. Vinyl
- 2. Phenyl
- 3. Methyl
- 4. Nitro

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.48 Read the passage carefully and answer the Questions

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pK_a of chloroacetic acid is lower than acetic acid because of

- 1. mesomeric effect
- 2. Resonance effect
- 3. due to -I effect of Cl atom and +I effect of methyl group.
- 4. Due to Hyperconjugation

Options 1. 1

2. 2

3. 3

4. 4

Q.49 Read the passage carefully and answer the Questions

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How can we express the (acidity constant) K_a of RCOOH ?

$$1. K_a = \frac{[\text{RCOO}^-][\text{RCOOH}]}{[\text{H}_3\text{O}^+]}$$

$$2. K_a = \frac{[\text{RCOO}^-][\text{H}_3\text{O}^+]}{[\text{RCOOH}]}$$

$$3. K_a = \frac{[\text{H}_3\text{O}^+]}{[\text{RCOOH}][\text{RCOO}^-]}$$

$$4. K_a = \frac{1}{[\text{RCOOH}][\text{RCOO}^-]}$$

Options 1. 1

2. 2

3. 3

4. 4

Q.50 Read the passage carefully and answer the Questions

Carboxylic acids are the earliest organic compounds isolated from nature. These are also known as organic acids but are weaker acids as compared to mineral acids. However, they are more acidic than alcohols and phenols because of greater resonance stabilization of the parent acid and its conjugate base. The acidic strength is determined by the K_a , which is called the acidity constant. The greater the value of K_a , the greater the tendency of the acid to ionize and hence the stronger the acid. The acidic strength of the saturated aliphatic carboxylic acid depends upon the inductive effect and its position with respect to $-\text{COOH}$ groups. Electron -donating substituents tend to decrease whereas electron -withdrawing substituents tend to increase the acidic strength. The acidic strength of the aromatic carboxylic acid, on the other hand depends upon both the inductive effect and the resonance effect of the substituents.

p-Nitrobenzoic acid is a stronger acid than benzoic acid because of

1. -R and -I effect of nitro group
2. Only -R effect of Nitro group
3. Only -I effect of Nitro group
4. +R effect and -I effect of nitro group

Options 1. 1

2. 2

3. 3

4. 4